

CORDILLERA FIRE & SAFETY SYSTEMS

Field Operations Safety & Protocols Handbook

Mining Fire Detection, Suppression & Emergency Response

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Revision history

Rev	Date	Summary of change
7.1	01 Apr 2026	Added lithium-ion / battery-electric equipment protocol (Section 12); updated PFAS transition guidance
7.0	15 Oct 2025	Aligned mobile-equipment provisions to AS 5062:2016 wording; revised inspection intervals.
6.2	20 Mar 2025	Updated NOM references; added refuge-alternative servicing checklist.
6.1	05 Aug 2024	Introduced CordilleraConnect monitoring-center protocol.

1. Purpose and Scope

This Handbook defines the mandatory safety policies, technical standards, and operating protocols that govern all Cordillera Fire & Safety Systems field activity at client mine sites in Mexico, including the design verification, installation, inspection, maintenance, monitoring, and emergency support of fire detection and suppression systems on mobile mining equipment and fixed mine infrastructure.

It applies to every Cordillera employee and subcontractor who performs work at, or remotely supports, a client site: field service technicians, monitoring-center operators, compliance auditors, installation crews, and supervising engineers. Compliance is a condition of employment. Where a client site standard, a Cordillera standard, and a legal requirement differ, the most stringent requirement applies.

This Handbook does not replace site-specific safety rules issued by the mine operator. Technicians must complete each client's site induction and obey all site rules in addition to this Handbook.

2. Safety Policy Statement

Cordillera exists because fire on mining equipment kills people and destroys assets. Our first obligation is that every person who works near our systems, and every Cordillera employee who installs and services them, goes home safe. We will never allow commercial or schedule pressure to compromise the integrity of a life-safety system. A technician who stops work because a system, a vehicle, or a situation is unsafe will always be supported, never penalized. — R. Salgado Mireles, Director General.

3. Regulatory and Standards Framework

Cordillera work is governed by Mexican law as the mandatory floor, and by recognized international standards adopted by clients as the engineering benchmark.

3.1 Mexican regulation (mandatory floor)

- **STPS Normas Oficiales Mexicanas (NOMs)** issued by the Secretaría del Trabajo y Previsión Social, governing safety conditions in underground and surface mining, workplace fire prevention and protection, handling of hazardous substances, and personal protective equipment.
- Mine-operator obligations under the applicable NOMs for ventilation, emergency response, evacuation, refuge provision, and fire prevention, which Cordillera systems are designed to help satisfy.
- Documented evidence of compliance must be maintained and made available to STPS inspectors and to the client's own audit function.

3.2 International standards (engineering benchmark)

- **AS 5062:2016** – Fire protection for mobile and transportable equipment. The de-facto global reference for mining mobile-equipment fire protection; defines the risk-assessment method, system design, agent quantities, detection, testing, and inspection/maintenance regime. Cordillera designs mobile-equipment systems to AS 5062 unless a client specifies a stricter standard.
- **NFPA 121** – Fire protection standard for self-propelled and mobile surface mining equipment.
- **NFPA 122** – Fire prevention and control in metal/nonmetal mines and processing facilities (fixed infrastructure).
- Component approvals (e.g., UL / FM listings) for cylinders, agents, detection, and actuation hardware.

Where a client has adopted an internal corporate standard derived from AS 5062 or NFPA, that internal standard is treated as the controlling engineering benchmark for that client's sites.

4. Fire Hazard Classification in Mining

Field personnel must be able to identify the dominant fire hazard before selecting or verifying a protection strategy. Cordillera classifies mining fire hazards as follows:

Class	Typical source	Primary control
Pressurized fluid	Hydraulic / fuel line rupture spraying onto hot surfaces (agent, suppression, PPE)	Isolation; line integrity; hot-surface shielding
Electrical	Battery, alternator, cabling, switchgear, electrical fires	Detection + suppression; isolation; gaseous/clean-agent for rooms
Conveyor / friction	Belt slip, roller seizure, material build-up underground	Linear heat detection cable; deluge / sprinkler; housekeeping
Diesel / hydrocarbon	Fuel, lube, hydraulic oil pooling and ignition	Dry chemical knockdown + liquid agent cooling; fuel-bay deluge
Lithium-ion / battery	Thermal runaway in battery-electric equipment	Early off-gas detection; containment; cooling (see Section 12)
Gas / atmosphere	Methane (coal), CO, oxygen deficiency	Gas detection; ventilation; refuge alternatives; evacuation

5. Equipment and System Standards

5.1 Suppression agents

- **Dry chemical** (mono-ammonium phosphate / sodium bicarbonate) for rapid flame knockdown. Does not cool; must not be relied on alone where re-ignition risk exists.
- **Liquid / wet-chemical agent** paired with dry chemical in **twin-agent** systems for cooling and securing against re-ignition. Twin-agent is the default for high-value mobile equipment.
- **Fluorine-free (F3) foam concentrate** for flammable-liquid hazards. PFAS-containing AFFF is being phased out (see Section 11); no new PFAS agent may be installed.
- **Clean / gaseous agents** for electrical and control rooms where powder or water damage is unacceptable.

5.2 Detection

- Spot heat detectors and **linear detection wire** in engine bays of mobile equipment.
- **Linear heat detection (LHD) cable** along conveyors, capable of locating the heat source along its length.
- Optical / infrared flame detectors for higher-risk mobile applications.
- Gas detection (CO, methane, O₂; battery off-gas) integrated to alarm and, where fitted, to CordilleraConnect.

6. Installation Protocol

No system may be commissioned until the following are complete and recorded:

- Risk assessment for the specific equipment type per AS 5062 (or the client's controlling standard) on file.
- Engineering design pack approved by a Cordillera supervising engineer (nozzle placement, agent type and quantity, detection layout, actuation).
- Components verified against the design pack and against approved/listed status; no substitution without engineering sign-off.
- Functional commissioning test witnessed and recorded.
- **As-built record** created and uploaded to the central asset register on the day of commissioning, including configuration, serial numbers, agent batch, and photographs. An installation is not closed until the as-built is in the system — this is mandatory and non-negotiable.
- Client hand-over pack issued and operator awareness briefing delivered.

Incomplete or missing as-built records are the leading cause of failed first-time fixes on later service visits and are treated as a quality non-conformance.

7. Inspection and Maintenance Protocol

Maintained systems save lives; unmaintained systems are decoration. Inspection intervals follow AS 5062 / the client standard and the system's duty:

Activity	Performed by	Typical interval
Operator pre-start visual check	Mine equipment operator	Each shift / daily
Technician periodic inspection	Cordillera field technician	Monthly to quarterly (by duty/risk)
Full functional / discharge test	Cordillera field technician	Per standard schedule (e.g., annual)
Agent and cylinder verification	Cordillera field technician	Each periodic inspection
Compliance audit vs. standard	Cordillera compliance auditor	Annual or on client request

- Every inspection must be recorded against the asset's as-built record. Verbal or undocumented work does not count as performed.
- Any system found impaired must be tagged, the client's responsible person notified in writing, and an interim risk control agreed before the equipment returns to service.
- Overdue inspections are tracked centrally; a protected asset whose inspection is overdue is escalated to the regional supervisor and the client.

8. Monitoring Center (CordilleraConnect) Protocol

The 24/7 monitoring center receives telemetry from connected systems and is the front line for real-time fire and fault events.

- Operators run three shifts providing continuous coverage every day of the year.
- On a **system activation (fire) signal**: immediately telephone the site's nominated contact, confirm personnel safety, log time/location/asset, and dispatch the nearest qualified technician for recharge and return-to-service.
- On a **health/fault signal** (low pressure, battery, detection-circuit fault): raise a service ticket, notify the site, and schedule corrective action within the contracted response window.

- All events are logged with timestamps; no event may be closed without a recorded resolution.
- Loss of telemetry from a connected asset is itself treated as a fault and investigated.

9. Emergency Response and Incident Management

- Cordillera personnel do not act as the mine's fire brigade. On discovering or being alerted to a fire, the priority order is: protect life, raise the alarm, evacuate/withdraw, and support the mine's emergency response only within training and PPE limits.
- Technicians must know each site's muster points, evacuation routes, refuge locations, and self-rescuer provisions before starting work.
- After any activation or fire, the system is treated as spent/unsafe until inspected, recharged, and re-commissioned by a qualified technician.

10. Refuge Alternatives and Self-Rescue Servicing

For underground operations, refuge alternatives and self-rescue equipment are life-critical and carry their own service obligations.

- Refuge chambers: verify seal integrity, independent air supply, CO₂/CO scrubbing media within shelf life, water, lighting, and communications at each scheduled service.
- Self-contained self-rescuers (SCSR): verify count, accessibility, and in-date status; remove and replace expired or damaged units.
- Scrubbing chemicals and breathing gas have finite shelf lives; expiry tracking is mandatory and recorded against each chamber's asset record.

11. Agent Handling and PFAS Transition

- No new PFAS-containing (AFFF) agent may be installed on any Cordillera-serviced system.
- Legacy AFFF encountered in the field must be logged; a fluorine-free (F3) transition plan is agreed with the client, and removed agent is disposed of through an approved waste route — never to ground or drains.
- F3 / wet-chemical agents may require different fill quantities and engineering; do not substitute agents without engineering sign-off.
- Handle dry chemical and concentrates with the PPE specified in the safety data sheet.

12. Lithium-Ion and Battery-Electric Equipment Protocol

Battery-electric mining equipment introduces a fundamentally different hazard. A lithium-ion fire can enter self-sustaining thermal runaway, reignite hours later, and release toxic/flammable gases; conventional knockdown agents alone are insufficient.

- Prioritize **early detection of battery off-gassing** ahead of visible flame, using gas detection integrated to alarm and to CordilleraConnect.
- Strategy emphasizes detection, containment, and cooling rather than knockdown alone; follow the equipment-specific engineered design.
- Treat a battery thermal event as potentially prolonged: maintain isolation distance, expect re-ignition, and do not assume a knocked-down fire is out.

- This is an evolving area; engineering issues equipment-specific bulletins as standards (including AS 5062 updates) and OEM guidance develop. Field personnel must use the current bulletin for each battery-electric asset.

13. Personal Protective Equipment and Lockout/Tagout

- Minimum site PPE: hard hat, eye protection, hi-vis, steel-toe boots, gloves, hearing protection, plus any client-mandated items (e.g., self-rescuer underground).
- Before working on any system, isolate stored energy: depressurize/secure cylinders, disable actuation, and apply lockout/tagout so a system cannot discharge during service.
- Never defeat a detection or actuation circuit without an agreed, time-bound impairment plan and client notification.

14. Training and Certification

- Every field technician must be trained and certified against this Handbook before working unsupervised, with documented refresher training on each revision.
- Competency is role-based: installation, periodic inspection, functional/discharge testing, refuge/SCSR servicing, and battery-electric work each require specific sign-off.
- Compliance auditors hold senior certification and audit client fleets against the controlling standard.
- Certification records are maintained centrally; an expired certification suspends the related field authorization.

15. Incident Reporting and Investigation

- Report all incidents, near-misses, and system non-conformances through the Cordillera reporting process within 24 hours; serious incidents immediately.
- Reportable items include any system that failed to activate or activated unintentionally, any injury, any impaired life-safety system, and any vehicle-road incident involving Cordillera travel.
- Investigations focus on system and process causes, not individual blame, and feed corrective actions and Handbook revisions.

16. Acknowledgement

By performing work governed by this Handbook, each employee and subcontractor confirms they have read it, understood it, hold current certification for the work they perform, and accept the authority — and the duty — to stop any work that is unsafe.

End of document. COR-HSE-HB-007 Rev 7.1. FICTIONAL training material — not a real organization or product.